

Experiment No. 3

Title: To Implement Sampling and Quantization

Apparatus use:

A PC with Python 3.7.8 and
OpenCV software Jupyter notebook

Theory:

Sampling and quantization

In order to become suitable for digital processing, an image function $f(x,y)$ must be digitized both spatially and in amplitude. Typically, a frame grabber or digitizer is used to sample and quantize the analogue video signal. Hence in order to create an image which is digital, we need to convert continuous data into digital form. There are two steps in which it is done:

1. Sampling
2. Quantization

The sampling rate determines the spatial resolution of the digitized image, while the quantization level determines the number of grey levels in the digitized image. A magnitude of the sampled image is expressed as a digital value in image processing. The transition between continuous values of the image function and its digital equivalent is called quantization.

- **Resize**

Syntax: cv2.resize(source, dsize, dest, fx, fy, interpolation)

Parameter:

source: Input Image array (Single-channel, 8-bit or floating-point)

dsize: Size of the output array

dest: Output array (Similar to the dimensions and type of Input image array)
[optional]

fx: Scale factor along the horizontal axis [optional]

fy: Scale factor along the vertical axis [optional]

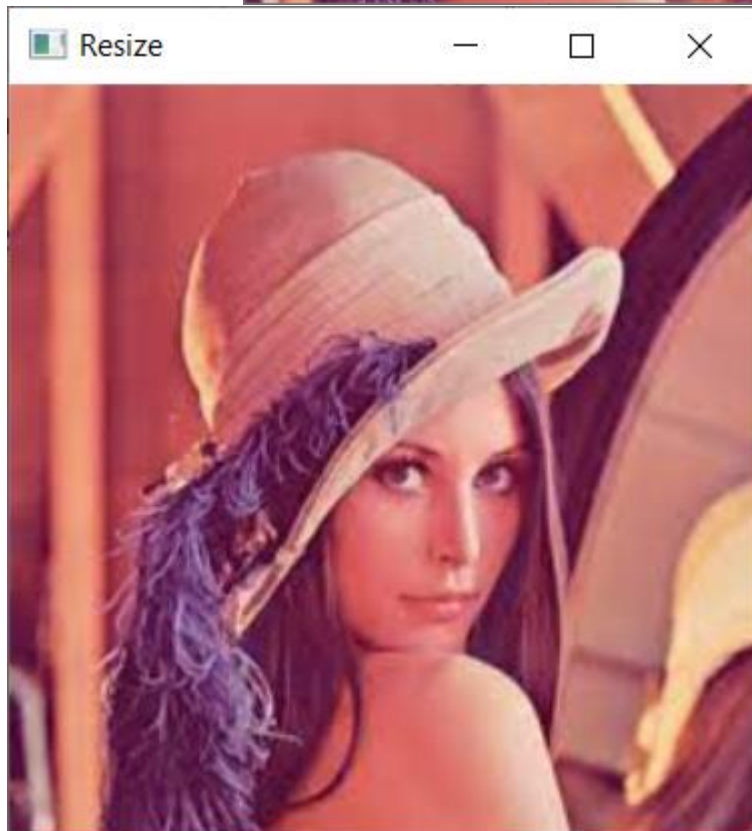
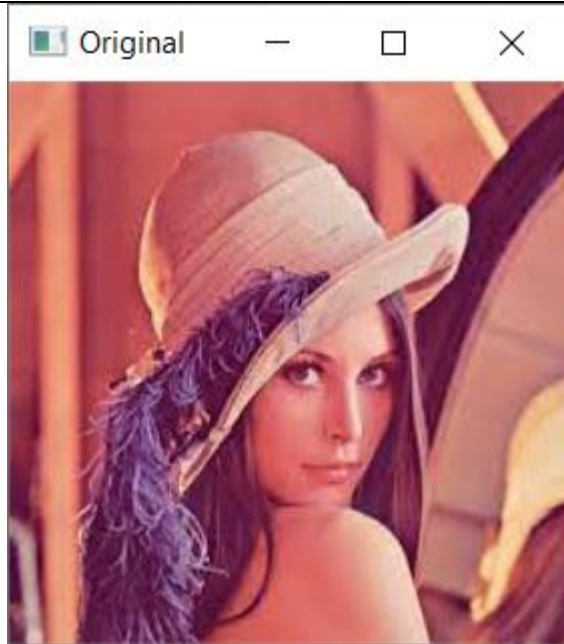
interpolation: One of the above interpolation methods [optional]

Code:

```
import cv2
src = cv2.imread('Lenna.jpeg')
img=cv2.resize(src, (300, 300))
print (src.shape)
print(img.shape)
cv2.imshow('Original',src)
cv2.imshow('Resize',img)
cv2.waitKey(0)
```

Output:

```
(156, 156, 3)
(300, 300, 3)
```



- **Resize using Percentage:**

Code:

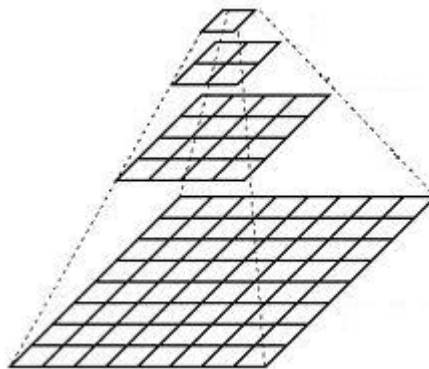
```
import cv2
src=cv2.imread("Lenna.jpeg")
src_shape=src.shape
height=src.shape[0]
```

```
width=src.shape[1]
percent=50
w = int(width * percent / 100)
h = int(height * percent / 100)
resize=cv2.resize(src,(h,w))
print(resize.shape)
cv2.imshow("Resize", resize)
cv2.waitKey(0)
```

- **Image Pyramid using OpenCV**

Image Pyramids are one of the most beautiful concepts of image processing. Normally, we work with images with default resolution but many times we need to change the resolution (lower it) or resize the original image in that case image pyramids comes handy.

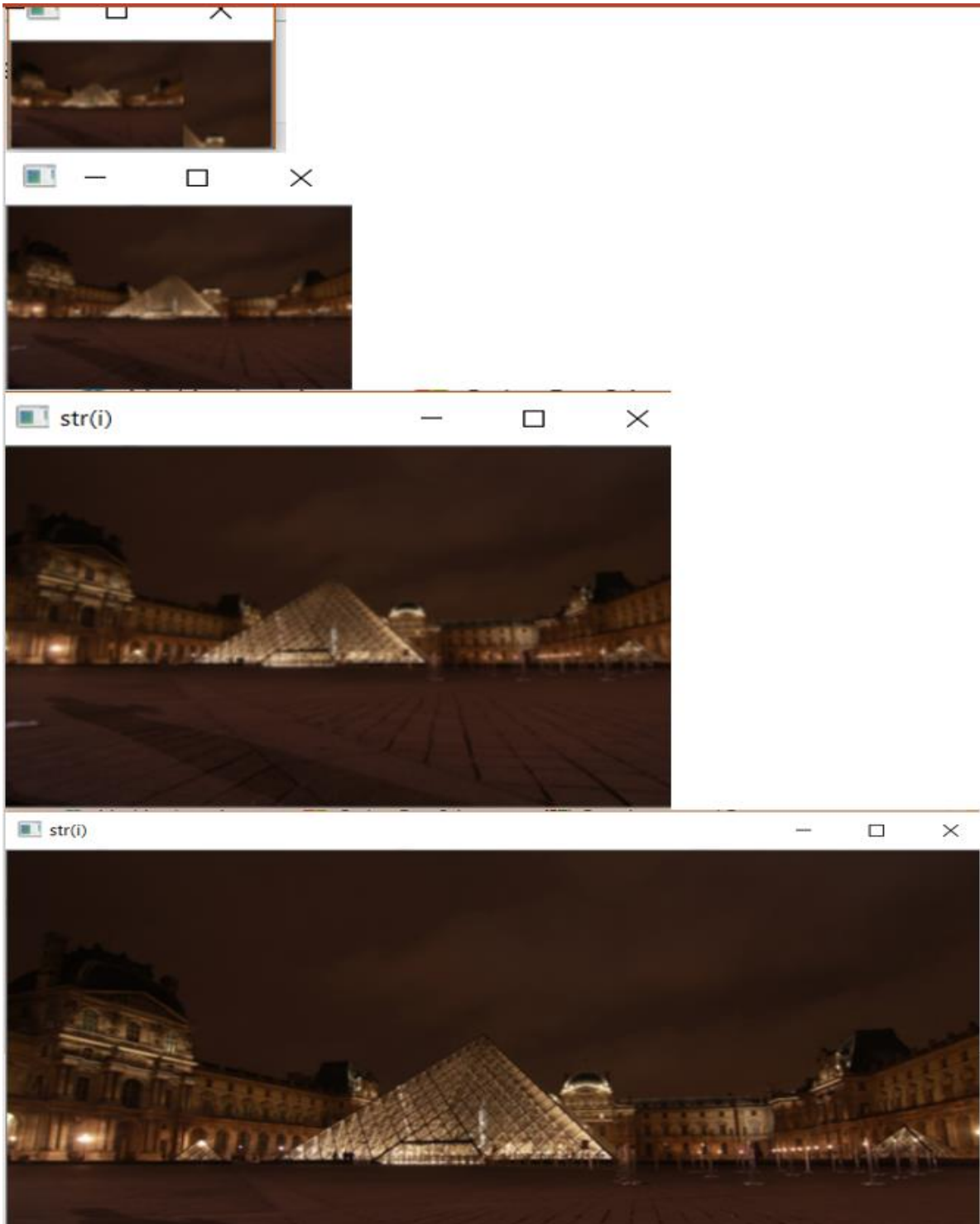
The **pyrUp()** function increases the size to double of its original size and **pyrDown()** function decreases the size to half. If we keep the original image as a base image and go on applying **pyrDown** function on it and keep the images in a vertical stack, it will look like a pyramid. The same is true for upscaling the original image by **pyrUp** function.



Code:

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("images/input.jpg")
layer = img.copy()
for i in range(4):
    plt.subplot(2, 2, i + 1)
    # using pyrDown() function
```

```
layer = cv2.pyrDown(layer)
plt.imshow(layer)
cv2.imshow("str(i)", layer)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output:

- **Image quantization**

PIL is the Python Imaging Library which provides the python interpreter with image editing capabilities. The Image module provides a class with the same name which is used to represent a PIL image. The module also provides a number of factory functions, including functions to load images from files, and to create new images.

Image.quantize() Convert the image to ‘P’ mode with the specified number of colors.

Syntax: Image.quantize(colors=256, method=None, kmeans=0, palette=None)

Parameters:

colors – The desired number of colors, <= 256

method – 0 = median cut 1 = maximum coverage 2 = fast octree

kmeans – Integer

palette – Quantize to the PIL.ImagingPalette palette.

Returns type: An Image object.

Code:

```
from PIL import Image
import PIL
img = Image.open(r"Lenna.jpeg")
img = img.quantize(170)
img.show()
```

Output:



Viva Questions:

- 1) What is sampling in the context of digital images?
- 2) What is quantization in image processing?
- 3) What is the difference between sampling and quantization?
- 4) Why are sampling and quantization necessary?
- 5) What happens to an image when you reduce the sampling rate?
- 6) What happens when you decrease the quantization level?
- 7) How do you choose the appropriate sampling rate?
- 8) What is the typical bit depth used for gray-scale images?
- 9) What interpolation techniques can be used during sampling?
- 10) What is meant by bit-depth in quantization?
- 11) What is the impact of high sampling rate with low quantization?
- 12) What is aliasing? How is it related to sampling?
- 13) How can you visually differentiate between an image affected by sampling vs quantization?
- 14) Why do we use gray-scale images for this experiment?
- 15) What metrics can you use to compare original and processed images?